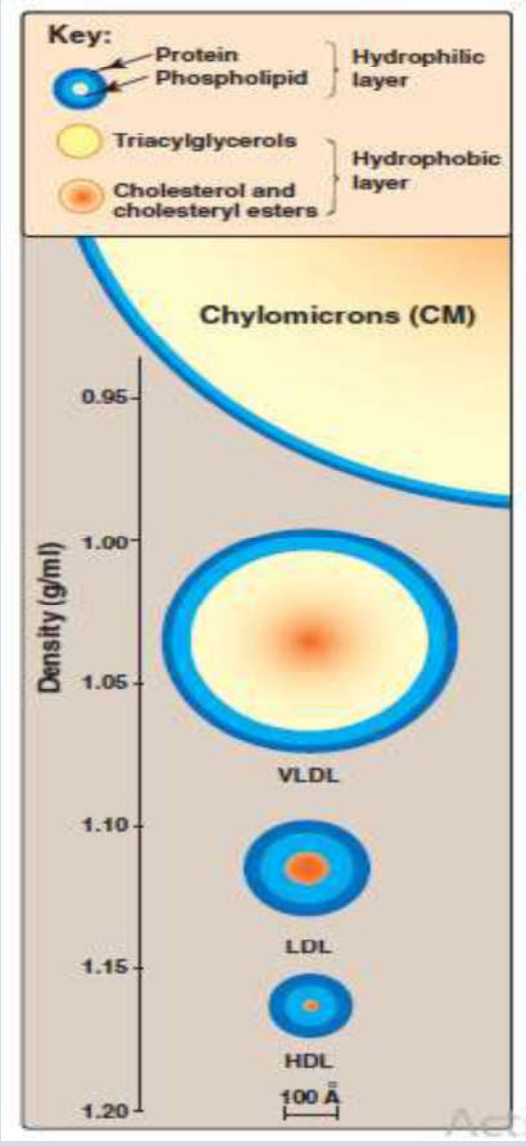
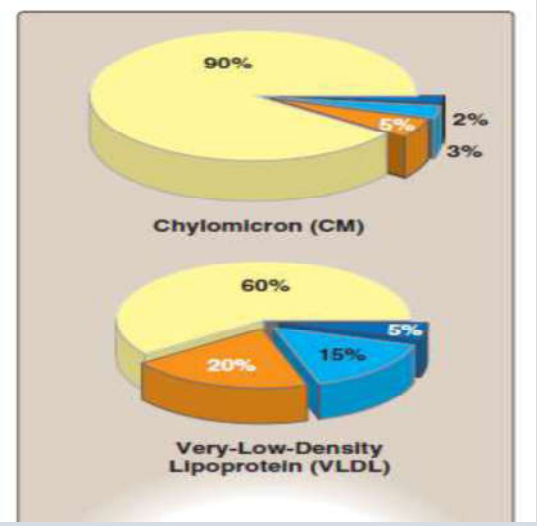
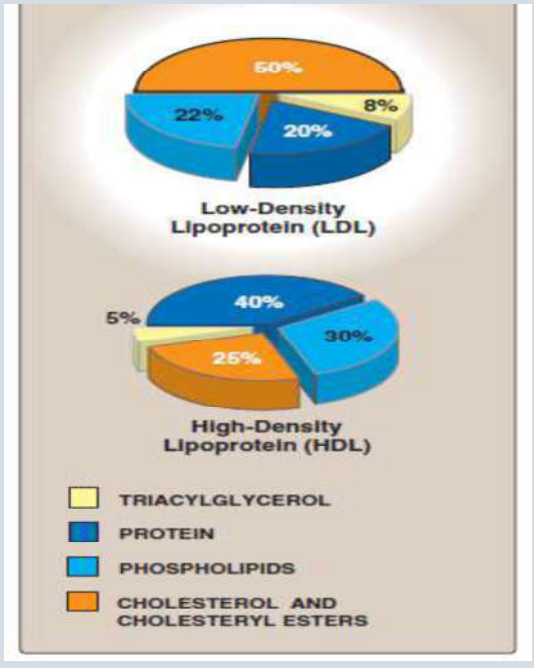
Lipoprotein

The plasma lipoproteins are spherical macromolecular complexes of lipids and specific proteins (apolipoproteins or apoproteins). The lipoprotein particles include chylomicrons (CM), very low density lipoproteins (VLDL), low-density lipoproteins (LDL), and high-density lipoproteins (HDL). They differ in lipid and protein composition, size, density, and site of origin. Lipoproteins function both to keep their component lipids soluble as they transport them in the plasma and to provide an efficient mechanism for transporting their lipid contents to (and from) the tissues. In humans, the transport system is less perfect than in other animals and, as a result, humans experience a gradual deposition of lipid especially cholesterol in tissues. This is a potentially life-threatening occurrence when the lipid deposition contributes to plaque formation, causing the narrowing of blood vessels (atherosclerosis).







Size and density of lipoprotein particles: Chylomicrons are the lipoprotein particles lowest in density and largest in size, and contain the highest percentage of lipid and the lowest percentage of protein. VLDLs and LDLs are successively denser, having higher ratios of protein to lipid. HDL particles are the densest. Plasma lipoproteins can be separated on the basis of their electrophoretic mobility, or on the basis of their density by ultracentrifugation.

Apolipoproteins: The apolipoproteins associated with lipoprotein particles have a number of diverse functions, such as providing recognition sites for cell-surface receptors, and serving as activators or coenzymes for enzymes involved in lipoprotein metabolism. Some of the apolipoproteins are required as essential structural components of the particles and cannot be removed (in fact, the particles cannot be produced without them), whereas others are transferred freely between lipoproteins. Apolipo proteins are divided by structure and function into five major classes, A through E, with most classes having subclasses, for example, apolipoprotein (or apo) A-I and apo C-II. [Note: Functions of all of the apolipoproteins are not yet known.]

CHYLOMICRONS metabolism:

Site of synthesis: intestinal mucosal cells

Functions: transport dietary lipids from intestine to peripheral tissues.

Structure:

a- Main lipids: triacylglycerols. Chylomicrons contain also cholesterol, phospholipids and fat soluble vitamins.

b- Proteins: (2%), apo B**48** and receives apo CII and apo E from **HDL**.

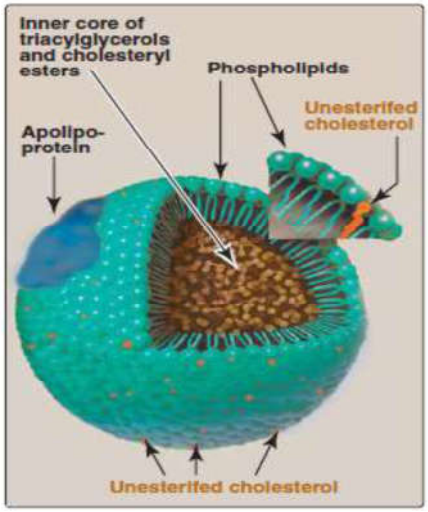
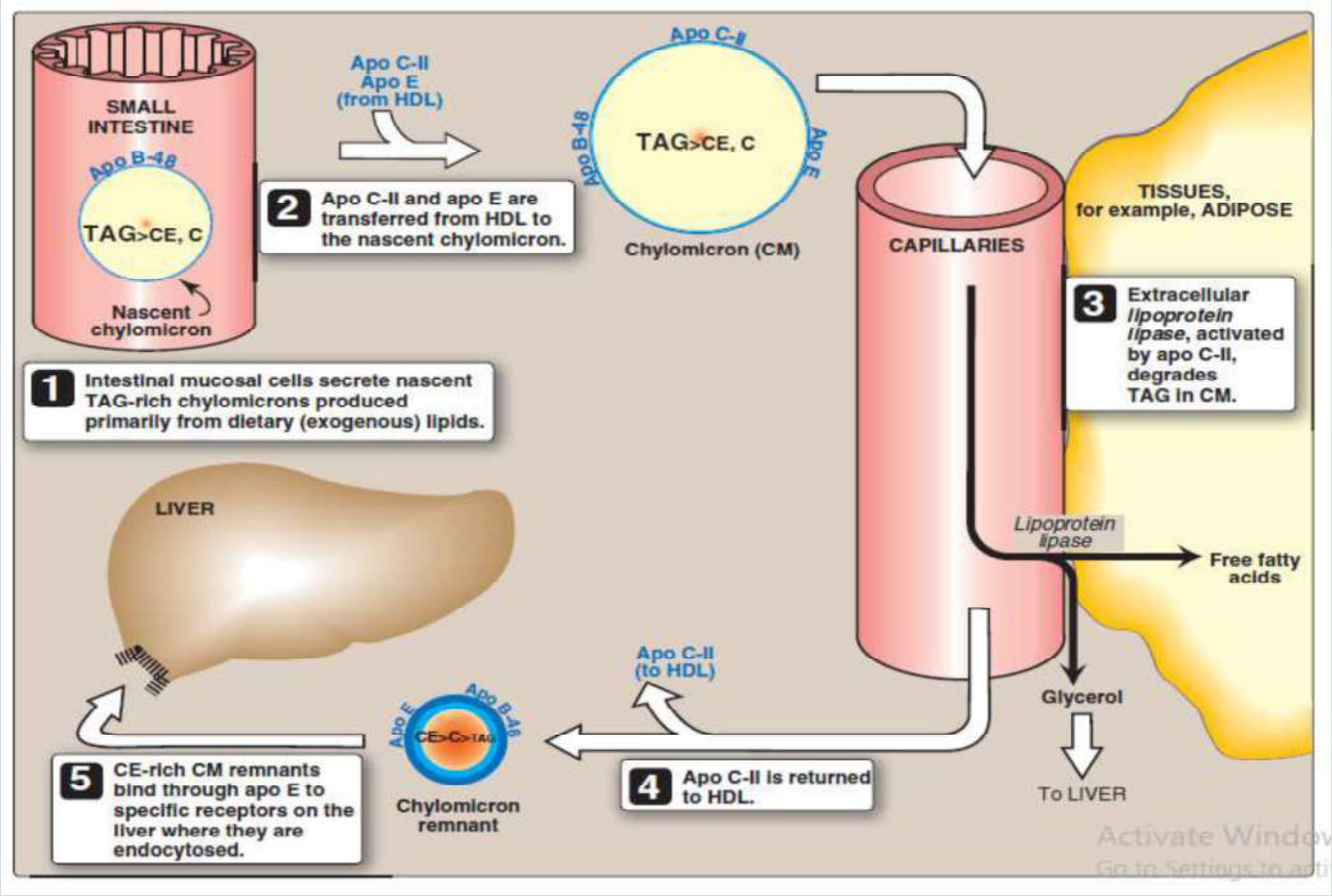


Figure 18.14
Structure of a typical lipoprotein particle.

Catabolism: TG are hydrolyzed by lipoprotein lipase (which is activated by apo CII). The remaining parts are **chylomicron remnants**, which are then taken up by the liver. Hepatocyte receptors can recognize apo B**48** and apo E.



VLDL Metabolism:

Site of synthesis: Liver.

Functions: transport lipids mainly TG from liver to peripheral tissues.

Structure:

a- Main lipids: triacylglycerols. It contains also cholesterol, phospholipids.

b- Proteins: (12%), apo B100 and receives apo CII and apo E from **HDL**.

Catabolism: TG are hydrolyzed by lipoprotein lipase (that is activated by apo CII). The remaining parts are IDL, which are then converted into LDL by transferring phospholipids, apo CII and apo E to HDL.

LDL Metabolism:

Site of synthesis: circulation from VLDL.

Functions: LDL particles provide cholesterol to peripheral tissues.

Structure:

a- Lipid contents: cholesterol, cholesterol esters and phospholipids.

b- Protein contents: (22%), apo B100

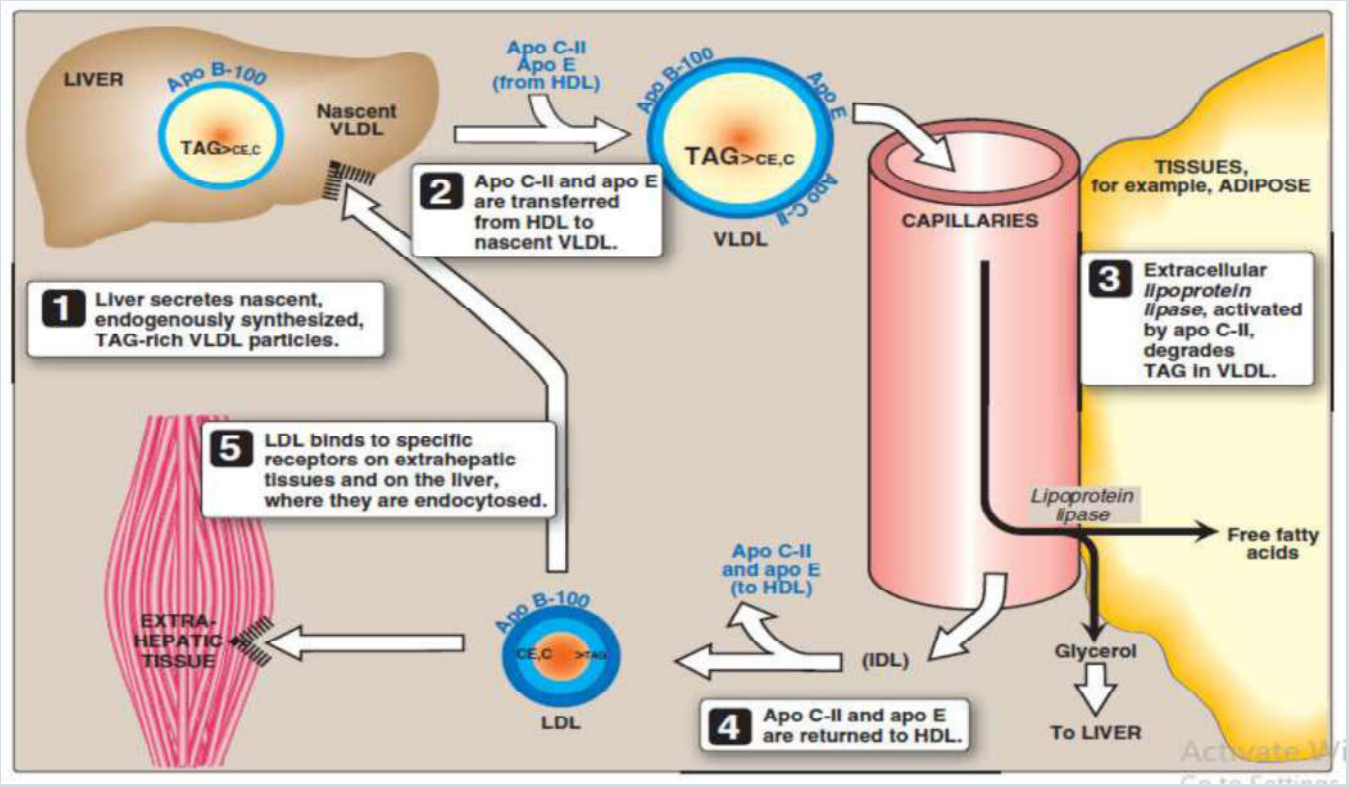
Catabolism:LDL apo B100 are recognized by tissue receptors. After binding with receptors, the LDL are internalized by endocytosis. Inside cells LDL are separated from receptors and hydrolyzed by lysosomal enzymes releasing cholesterol, amino acids, fatty acids and phospholipids.

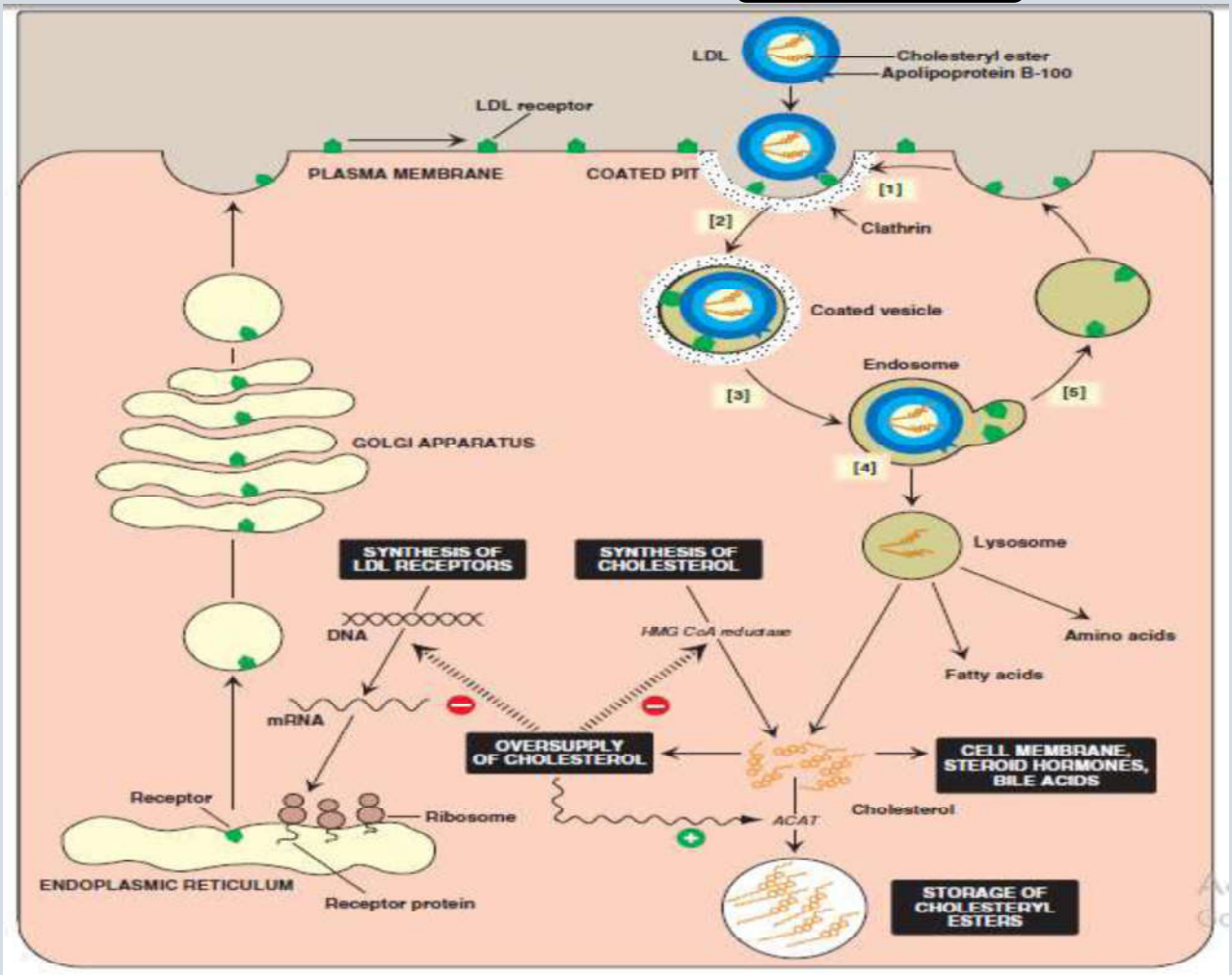
***If the cell contains oversupply of cholesterol from LDL, HDL or chylomicron remnants, the cholesterol amount can be decreased by:**

a- Inhibition of HMG-CoA reductase → Inhibition of cholesterol synthesis.

b- Stimulation of **ACAT** enzyme → Cholesterol ester.

c- Inhibition of synthesis of **LDL receptors** → inhibition of LDL uptake by cells.





HDL metabolism:

Site of HDL synthesis: Liver.

Functions:

a- Contain Apo CII, which activates **lipoprotein lipase** that hydrolyzes TG.

b- Remove cholesterol from peripheral tissues and esterified it by **LCAT** enzyme → Cholesterol esters.

c- Carry cholesterol esters to VLDL & chylomicrons to the liver.

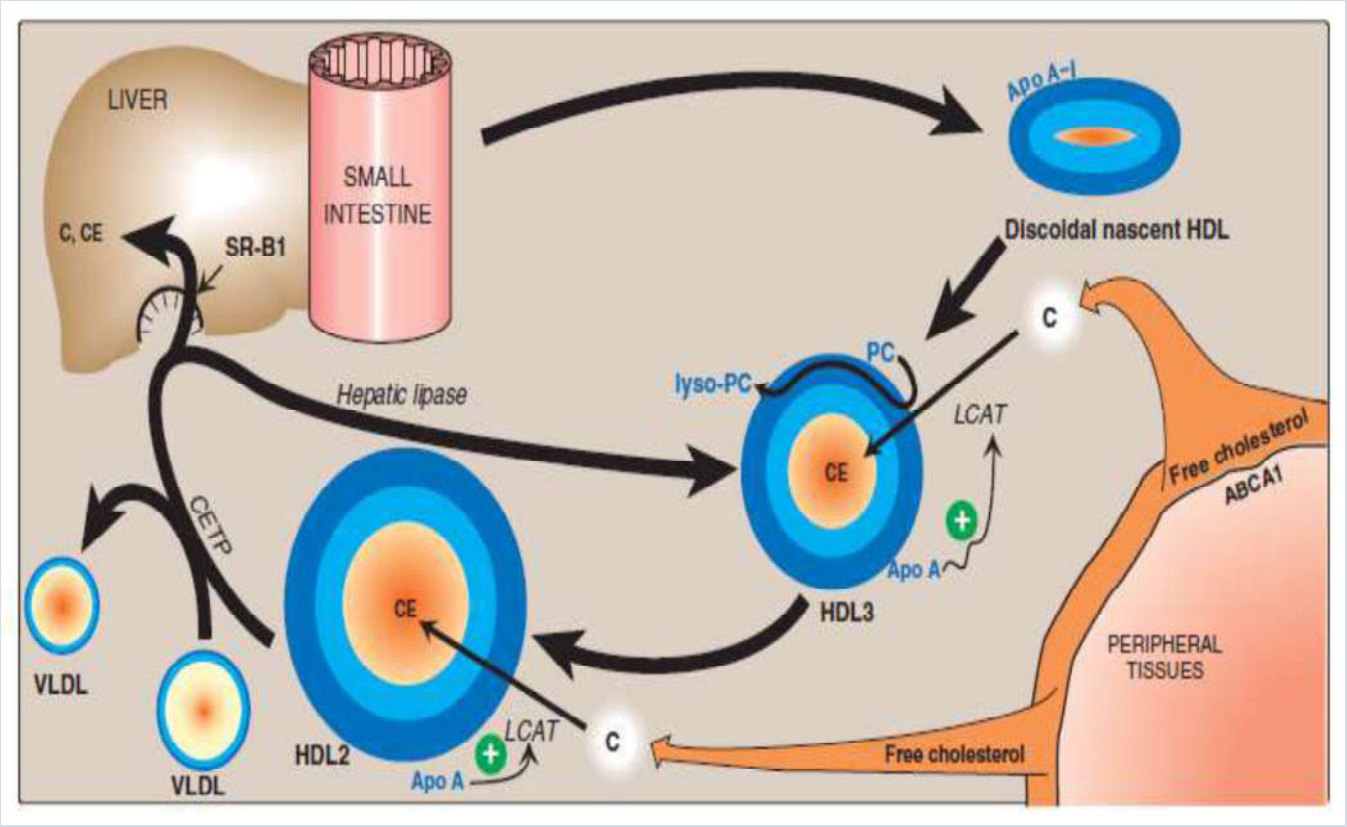
Structure:

a- Main lipids: cholesterol (free and esterified) and phospholipids, mainly lecithin.

b- Proteins: (50%), apo A-1 (which activates **LCAT**), apo CII (which activates **lipoprotein lipase** and E (which is recognized by hepatic receptors)

Catabolism: HDL accepts unesterified cholesterol from peripheral tissues. Then HDL particles (by **LCAT** enzyme) esterify cholesterol into cholesterol

esters. Then HDL is taken up by the liver cells where Cholesterol esters are released inside them to be utilized in the formation of lipoproteins or excreted in bile.



APOPROTEINS (APOLIPOPROTEINS)

Definition: These are proteins (globulins) present in association with plasma lipids to form lipoproteins.

Functions:

1. Apolipoproteins form with lipids water-soluble compounds, so they help transport of lipids between tissues.

2. Some apoproteins activate certain enzymes e.g. ***Apo C II*** activates lipoprotein lipase and ***Apo-A1*** activates LCAT.

3. They act as liganda (connection) for interaction of lipoprotein with their receptors in tissues i.e. receptors of lipoproteins in tissues can recognize lipoproteins through their apoproteins e.g. apo B100 for LDL receptors.

Classification

Classification

	LIPOPROTEIIIS	COMMEIIT
Apo A-I	HD L	*Activator of LCAT *Legend for HDL-receptors
Ape B- 100	LDL-YLDL-IDL	*synthesized by liver *Legend for LDL-receptcrs
Ape B- 48	Chylomicrons - chylomicron remnants	*Synthesized by intestine *Legend for chylomicron remnant receptors in Liver.
Ape C II	Chylomicrons-VLDL-HDL	Activator of lipoprotein lipase
Ape D	HDL	May act as lipid transfer protein.
Ape E	Chylomicrons - Chylomicron remnants – VLDL - HDL	* Legend for chylomicron remnant receptors in liver. * Legend for LDL receptors in tissues.

Activate V